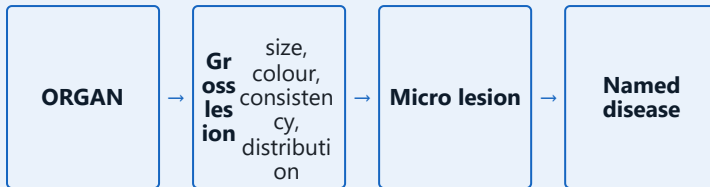


1 · HOW TO ANSWER ANY SYSTEMIC QUESTION



DISTRIBUTION WORDS — WORTH MARKS Focal (one) · Multifocal (many separate) · Locally extensive · Diffuse (whole organ) · Cranioventral (bronchopneumonia) · Random (haematogenous/embolic) · Centrilobular (liver, hypoxia) · Periacinar.

2 · MOUTH — STOMATITIS BY TYPE ★

Type	Diseases
Vesicular	FMD , vesicular stomatitis, vesicular exanthema, swine vesicular disease — the 4 look identical, only the lab separates them
Erosive / ulcerative	Rinderpest, BVD-MD, MCF, PPR , uraemia, bluetongue
Papular	Pox, contagious ecthyma (orf) — proliferative scabby lips of sheep/goats
Necrotic	Calf diphtheria — <i>Fusobacterium necrophorum</i> ; foul odour, yellow-grey necrotic membrane
Mycotic	Thrush — <i>Candida albicans</i> , white pseudomembrane

TWO JAWS — NEVER MIX THESE ★★ WOODEN TONGUE = *Actinobacillus lignieresii* → **SOFT tissue** (tongue), pyogranuloma with **club colonies / sulphur granules**.
LUMPY JAW = *Actinomyces bovis* → **BONE** (mandible), rarefying + proliferative osteomyelitis.
 Mnemonic: Actinob**A**cillus = tongue is a b**A**g of muscle;
 Actino**MYCES** = Mandible.

3 · FORESTOMACHS — THE BOVINE SET

- **Ruminal tympany (bloat)** — **Primary/frothy**: legume & lush green pasture, stable foam. **Secondary/free-gas**: eructation blocked (oesophageal obstruction, vagal indigestion).
- **Proof it was antemortem ★**: the **BLOAT LINE** — a sharp line in the cervical oesophagus, **congested below** the thoracic inlet and **pale above**, plus pale, bloodless head and forelimbs and a distended abdomen.
- **Rumenitis–liver abscess complex** — grain overload → **lactic acidosis** → ruminal ulcer → portal spread of *Fusobacterium necrophorum* → **liver abscess** → may rupture into the caudal vena cava.
- **Traumatic reticuloperitonitis ("hardware disease")** — wire through the reticulum → localised peritonitis; forward → **TRAUMATIC PERICARDITIS**: thick foul "washing-water" pus and fibrin, the classic bovine heart.
- **Parakeratosis of the rumen** — hard concentrate diet; dark, clumped, hardened papillae.

4 · STOMACH · ABOMASUM · INTESTINE

- **Ostertagiosis** — abomasal nodules give the "**Morocco leather**" appearance; ↑ abomasal pH, hypoproteinaemia, **bottle jaw**
- **Abomasal displacement (LDA/RDA) & torsion** — high-yielding dairy cows
- **Gastric ulcer** — pigs (pars oesophagea), stress, NSAIDs, uraemia, mast cell tumour in dogs

Enteritis	Classic cause
Catarrhal	Mild viral, dietary
Haemorrhagic	Canine parvovirus , coccidiosis, enterotoxaemia , anthrax
Fibrinous / diphtheritic	Salmonellosis , classical swine fever (button ulcers in caecum/colon), rinderpest
Necrotic	Salmonellosis, clostridial
Granulomatous ★	JOHNE'S DISEASE — corrugated, rugose, thickened ileum that will not flatten, cording of lymphatics , enlarged mesenteric nodes, NO ulceration
Eosinophilic	Parasites, allergy

Displacements: intussusception (telescoping — invaginated segment is the intussusceptum) · volvulus · torsion · hernia · atresia ani.

5 · LIVER & PANCREAS

- **Hepatitis** — necrotising (ICH, salmonellosis), suppurative (abscess), chronic (cirrhosis)
- **CIRRHOSIS** — **diffuse fibrosis + regenerative nodules + architectural distortion**; hard, shrunken, nodular liver. Causes: aflatoxin, pyrrolizidine alkaloids (*Crotalaria*, *Senecio*), copper, chronic fasciolosis, chronic passive congestion (cardiac cirrhosis)
- **"Pipe-stem liver"** — thick fibrotic, calcified bile ducts of chronic **fasciolosis** (cattle); **black pigment** = parasitic haematin
- **Telangiectasis** — dark blood-filled spaces in cattle liver, incidental
- **"Sawdust liver"** — pale foci of focal necrosis in feedlot cattle
- **Hepatositis dietetica** — pig, **vitamin E / selenium deficiency**; mosaic pattern of lobular necrosis
- **Nutmeg liver** — chronic passive congestion (right heart failure)
- **Pancreas** — acute **necrotising pancreatitis** with **chalky-white fat necrosis** (dog); pancreatic atrophy → EPI, steatorrhoea

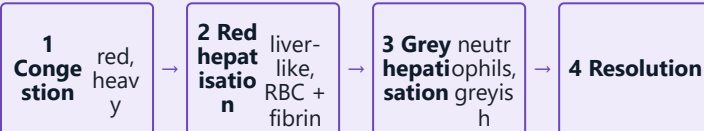
ICTERUS IN INDIA Yellow carcass + dark urine + splenomegaly in a crossbred cow in tick season = **babesiosis/theileriosis**, not liver disease.

6 · RESPIRATORY — THE 4 PNEUMONIA PATTERNS ★★★

Pattern	Distribution	Route	Examples
Broncho-pneumonia	CRANIOVENTRAL , lobular, sharply demarcated	Aerogenous (airway)	Suppurative/catarrhal — <i>Pasteurella</i> , <i>Mannheimia</i> , mycoplasma. Fibrinous (pleuropneumonia) — HS, shipping fever, CBPP
Interstitial	DIFFUSE , whole lung, rubbery , fails to collapse, rib impressions	Aerogenous or haematogenous	Viral (PPR, RD, IBR, CDV), toxic, 3-methylindole atypical interstitial pneumonia (fog fever)
Embolic	MULTIFOCAL, RANDOM , all lobes	Blood-borne	Ruptured liver abscess, vegetative endocarditis, navel ill
Granulomatous	Multifocal nodules	Both	Tuberculosis , mycose, aspiration of foreign material

SIGNATURE LUNG ★ "MARBLING" = massive **interlobular septal thickening** + hepatisation → **CBPP**. **Sequestra** in a marbled lung = chronic CBPP.

7 · LUNG — THE OTHER TERMS



- **Atelectasis** — airless collapsed lung; **congenital/neonatal** (never breathed, sinks in water) vs **acquired** (compression / obstruction, absorption)
- **DOCIMASIA (flotation) test ★**: lung that has breathed **FLOATS**; unbreathed fetal lung **SINKS** — used in veterolegal stillbirth cases
- **Emphysema** — **alveolar** (destruction of septa) and **interstitial** (air in interlobular septa; classic in **cattle**, may track to the back as subcutaneous emphysema)
- **Atrophic rhinitis of pigs** — toxigenic *Pasteurella multocida* type D + *Bordetella bronchiseptica* → **turbinate atrophy**, **snout deviation**
- **Verminous pneumonia** — *Dictyocaulus*, cranioventral wedge-shaped grey firm areas
- **Aspiration pneumonia** — right cranial/middle lobes, foul, gangrenous; drenching accident is the classic cause in India

8 · MUSCLE

Condition	Key lesion
White muscle disease (nutritional muscular dystrophy)	Vitamin E / selenium deficiency ; calves, lambs. Chalky white streaks in skeletal muscle and heart; Zenker's (hyaline) degeneration + dystrophic calcification. ↑ CK & AST . Bilaterally symmetrical
Blackleg	<i>Cl. chauvoei</i> ; dark, dry, spongy, crepitant muscle with a rancid butter odour; heavy muscle of young well-fed cattle
Malignant oedema	<i>Cl. septicum</i> , wound entry, more oedematous and less emphysematous than blackleg
Azoturia (exertional rhabdomyolysis)	Horse after rest on full grain then work; gluteal muscles hard & painful, coffee-coloured myoglobinuric urine
Capture myopathy	Wild animals after chase/restraint — Zenker's necrosis, myoglobinuria
Eosinophilic myositis	<i>Sarcocystis</i> ; green streaks in beef, a meat-inspection condemnation

MYOGLOBIN VS HAEMOGLOBIN ★ Myoglobinuria — plasma stays **clear** (myoglobin is small, cleared fast). **Haemoglobinuria** — plasma is **pink/red**. This single point separates azoturia from babesiosis and post-parturient haemoglobinuria.

9 · BONE & JOINT ★

Disease	Cause & lesion
Rickets	YOUNG, growing animal. Vitamin D / phosphorus deficiency. Failure of mineralisation of osteoid at the growth plate → wide irregular physis, enlarged costochondral junctions = " rachitic rosary ", bowed limbs
Osteomalacia	ADULT — same defect after the growth plates close; soft bones, pathological fracture, pica
Fibrous osteodystrophy	Nutritional secondary hyperparathyroidism — low Ca : high P diet (bran, all-grain). Bone replaced by fibrous tissue → " big head " of horses (bran disease); " rubber jaw " in renal secondary hyperparathyroidism of dogs
Osteoporosis	Normal mineralisation but too little bone ; starvation, Ca deficiency, disuse
Osteomyelitis	Haematogenous in young; sequestrum (dead bone) + involucrum (new bone sheath)
Fluorosis	Mottled, pitted, chalky teeth; exostoses on mandible & metatarsals; lameness

Joints: suppurative arthritis = joint-ill / navel-ill of calves and foals (*Streptococcus*, *E. coli*, from failure of passive transfer) · **degenerative joint disease/osteoarthritis** — cartilage erosion, eburnation, **osteophytes** · **osteocondrosis dissecans** — flap of cartilage in fast-growing large breeds.

10 · HEART — PERICARDIUM, MYOCARDIUM, ENDOCARDIUM

Layer	Lesion and the disease it names
Pericardium	Fibrinous pericarditis = " bread-and-butter " heart. Traumatic pericarditis of cattle — foul "washing-water" pus + wire. Hydropericardium — mulberry heart disease of pigs (vit E/Se) and hydropericardium syndrome of broilers
Myocardium	"TIGER HEART" ★ — pale yellow-white streaks alternating with red on the papillary muscles: FMD in calves (also white muscle disease). Myocarditis in blackleg, trypanosomosis, CSF
Endocardium	Vegetative valvular endocarditis — cauliflower-like friable masses. Cattle: right AV (tricuspid) ; pig: left AV (mitral) with <i>Erysipelothrix</i> and <i>Streptococcus</i> . Source of embolic nephritis and pneumonia

HYPERTROPHY TYPE CONCENTRIC — pressure overload, thick wall, **small lumen** (aortic/pulmonic stenosis). **ECCENTRIC** — volume overload, dilated chamber, wall relatively thin (valvular insufficiency). **Cor pulmonale** = right heart hypertrophy from lung disease; **brisket disease** = high-altitude cor pulmonale of cattle.

11 · BLOOD VESSELS

- **Arteriosclerosis** — generic hardening; **atherosclerosis** (lipid plaque) is rare in animals but occurs in **hypothyroid dogs** and pigs
- **Verminous aneurysm ★** — *Strongylus vulgaris* in the **cranial mesenteric artery** of the horse → thrombosis → intestinal infarction and colic
- **Arteritis** — equine viral arteritis, malignant catarrhal fever, immune-complex (Type III) vasculitis
- **Lymphangitis** — epizootic lymphangitis (*Histoplasma farciminosum*) and glanders (**farcy**) in horses; corded lymphatics in **Johne's**

12 · ANAEMIA — CLASSIFY IT TWO WAYS ★★

By cause	Examples
Blood loss (regenerative)	Haemonchosis , ticks, lice, hookworm, trauma, ulcer
Haemolytic (regenerative)	Babesiosis , theileriosis , anaplasmosis , leptospirosis, copper toxicity, neonatal isoerythrolysis, AIHA
Reduced production (non-regenerative)	Bracken fern , radiation, chronic renal failure (↓erythropoietin), iron deficiency, chronic inflammation, trichlorethylene-extracted soybean

By RBC morphology	Meaning
Macrocytic hypochromic	Regenerative — young cells released; reticulocytosis, polychromasia, anisocytosis, nucleated RBCs, Howell-Jolly bodies
Normocytic normochromic	Acute loss, anaemia of chronic disease, early stage
Microcytic hypochromic	IRON DEFICIENCY — classically baby-pig anaemia and chronic blood loss

SPECIES TRAP The **HORSE** does **NOT** release **reticulocytes** into blood, so equine anaemia cannot be called regenerative on a blood smear — you must examine **bone marrow**.

13 · BONE MARROW & WHITE CELLS

- **Aplastic anaemia / pancytopenia** — **bracken fern** (ptaquiloside), radiation, chloramphenicol, oestrogen toxicity in dogs, trichothecene mycotoxins
- **Myelophthisis** — marrow crowded out by tumour
- **Leukaemia** — neoplastic cells **in blood**; **lymphoma/lymphosarcoma** = solid tumour of lymphoid tissue, may be **aleukaemic**
- **Polycythaemia** — relative (dehydration, the common one) vs absolute (primary or hypoxia-driven)
- **Leucopenia** — **viral** (CSF, BVD, IBD, panleukopenia, RD), overwhelming sepsis, endotoxaemia, radiation

14 · LYMPH NODE, SPLEEN, THYMUS, BURSA

- **Acute lymphadenitis** — enlarged, wet, bulging, red
- **Caseous lymphadenitis** — *Corynebacterium pseudotuberculosis*, sheep/goats; **concentric "ONION-RING"** laminated pus
- **Lymphoid atrophy/depletion** — **IBD (Gumboro)** in the bursa, BVD, CDV, stress and corticosteroid

Spleen	Meaning
"Blood spleen" ★	ANTHRAX — grossly enlarged, soft, dark, blackberry-jam / tarry pulp that oozes
Sago spleen	Nodular (follicular) amyloidosis
Lardaceous spleen	Diffuse red-pulp amyloid
Splenic infarct	Classical swine fever — marginal wedge-shaped dark infarcts (near pathognomonic)
Siderotic plaques	Brown-yellow iron-calcium plaques on the capsule of old dogs, incidental

ANTHRAX RULE ★★★ A ruminant found dead with **tarry unclotted blood from body orifices**, **rigor mortis absent or incomplete**, and a huge soft spleen — **DO NOT OPEN THE CARCASS**. Take a **peripheral blood smear** (ear vein), stain with **polychrome methylene blue** → **McFadyean's reaction** (pink capsule around blue rods), and dispose by **deep burial with lime or incineration**.

15 · KIDNEY — WHERE THE LESION SITS ★★★

Disease	Site	Signature
Glomerulonephritis	Glomerulus	Immune complex (Type III); proteinuria, protein-losing nephropathy , oedema/ascites
Interstitial nephritis	Interstitium	"WHITE-SPOTTED KIDNEY" of calves — leptospirosis , <i>E. coli</i> ; grey-white foci with lymphocyte cuffs
Embolic suppurative nephritis	Cortex, random	Multiple small abscesses from valvular endocarditis or navel-ill
Pyelonephritis	Pelvis → medulla, ascending	<i>Corynebacterium renale</i> in cattle; pus in the pelvis, radiating streaks in medulla
Nephrosis (tubular necrosis)	Tubules	Toxic — oak/tannin, oxalate , aminoglycoside, ethylene glycol (birefringent calcium oxalate crystals), mercury, ochratoxin, sulphonamide
Amyloidosis	Glomerulus	Congo red, apple-green birefringence ; pale, waxy, enlarged kidney
Hydronephrosis	Whole kidney	Obstruction → pressure atrophy , kidney becomes a thin-walled sac

URAEMIA — THE LESIONS EXAMINERS WANT Ulcerative stomatitis & glossitis · uraemic gastritis with mucosal mineralisation · **fibrinous pericarditis** · pulmonary oedema ("uraemic lung") · **intercostal & aortic mineralisation** · **rubber jaw** in dogs · ammoniacal breath.

16 · LOWER URINARY TRACT

- **Urolithiasis** — **struvite** (Mg-ammonium-phosphate; concentrate-fed male sheep, goats and bullocks) · **silica** (range cattle) · **oxalate** · urate (Dalmatian). Obstruction at the **sigmoid flexure / urethral process** → **"water belly"** (urethral rupture, urine in subcutis) or bladder rupture and uroperitoneum
- **Cystitis** — ascending, commoner in females
- **ENZOOTIC HAEMATURIA ★** — chronic **bracken fern** feeding → bladder **haemangiomas and carcinomas**, persistent haematuria; often with **bovine papillomavirus** co-factor

17 • FEMALE GENITALIA & PLACENTA

- **Endometritis / metritis** — post-partum, retained placenta, dystocia; *Trueperella* (*Arcanobacterium*) *pyogenes*, *E. coli*
- **PYOMETRA** — **cystic endometrial hyperplasia–pyometra complex** of the bitch, **progesterone**-driven, ***E. coli***; **closed** cervix is the dangerous form (endotoxaemia, ↑ risk of rupture)
- **BRUCELLA placentitis** ★ — **necrotic, yellow, leathery cotyledons** + **thickened intercotyledonary membrane** with a sticky exudate; fetal stomach content is the best sample
- **Mummification** (fetus dries, no bacteria, closed cervix) vs **maceration** (bacteria enter, soft tissues liquefy, bones remain)
- **Cystic ovarian disease** — follicular cyst (nymphomania) vs luteal cyst (anoestrus)
- **Hydrallantois / hydramnios, uterine torsion, prolapse**

ABORTION CAUSES TO NAME **Bacterial:** brucellosis, leptospirosis, listeriosis, campylobacteriosis, salmonellosis, *Chlamydia*. **Viral:** IBR, BVD, EHV-1 (rhinopneumonitis), bluetongue, PPR. **Fungal:** *Aspergillus* — raised leathery plaques on the placenta and **ringworm-like skin lesions on the fetus**. **Protozoal:** *Neospora*, *Toxoplasma* (sheep — **white necrotic foci on the cotyledons**, "strawberry jam"). **Toxic/plant:** nitrate, locoweed.

18 • MALE GENITALIA & UDDER

- **Orchitis** — *Brucella abortus* in bulls, *B. ovis epididymitis* in rams (tail of the epididymis); leads to **testicular degeneration and sterility**
- **Cryptorchidism** — retained testis; **higher risk of Sertoli cell tumour** and **seminoma**; spermatogenesis fails but Leydig cells still make testosterone
- **Balanoposthitis** — IBR/IPV, *Corynebacterium renale* (pizzle rot of wethers)

Mastitis	Agent / lesion
Catarrhal / subclinical	<i>Streptococcus agalactiae</i> ; ↑ somatic cell count, CMT positive
Suppurative	<i>Staph. aureus</i> , <i>Str. dysgalactiae</i> ; abscess, fibrosis, "blind quarter"
GANGRENOUS ★	<i>Staph. aureus</i> (α-toxin) & <i>Cl. perfringens</i> ; udder cold, blue-black, sloughs; blood-tinged watery secretion
Summer mastitis	<i>Trueperella pyogenes</i> , fly-borne, dry cows, foul thick pus
Granulomatous	Tuberculous mastitis — a public-health hazard in India; also <i>Nocardia</i> , mycotic

19 • NERVOUS SYSTEM — REACTION PATTERNS

Chromatolysis Nissl loss, swollen neuron

Satellitosis glia around a neuron

Neuronophagia dead neuron eaten

Gliosis the CNS scar

Perivascular cuffing = **VIRAL**

Status spongiosus vacuolated neuropil

Wallerian degeneration

NAMING THE SITE **Polioencephalitis** = **GREY** matter (rabies, RD, listeriosis) · **Leukoencephalitis** = **WHITE** matter (CDV, scrapie) · **malacia** = softening/necrosis · **Meningitis** · **Myelitis** = spinal cord.

Named CNS disease	Cause & lesion
Polioencephalomalacia (CCN)	Thiamine deficiency / thiaminase, high grain, amprolium, high sulphur. Cortex soft, yellow, laminar necrosis; cut surface AUTOFLUORESCES under UV ; opisthotonus, blindness
Leukoencephalomalacia	Mouldy maize , <i>Fusarium moniliforme</i> fumonisin , HORSES ; liquefactive cavities in cerebral white matter
Nigropallidal encephalomalacia Rabies	Yellow star thistle, horses — "chewing disease" NEGRI BODY — intraCYTOplasmic eosinophilic inclusion in hippocampal pyramidal neurons and cerebellar Purkinje cells ; non-suppurative polioencephalitis. Stain: Seller's
Listeriosis	" Circling disease "; microabscesses in the BRAIN STEM (pons and medulla) , unilateral cranial nerve signs; silage-fed
Sway back / enzootic ataxia	COPPER deficiency in lambs — cerebral white-matter cavitation and spinal demyelination
Cerebellar hypoplasia	BVD in calves, feline panleukopenia in kittens — in-utero infection
Hepatic encephalopathy	Status spongiosus + Alzheimer type II astrocytes ; ammonia

20 • ENDOCRINE

Gland	Disorder
Thyroid	GOITRE — iodine deficiency or goitrogens (brassicas, soybean, thiocyanate); hyperplastic (young, colloid-poor) vs colloid goitre. Newborns: weak, hairless, stillborn with a swollen neck . Hyperthyroidism — old cats, functional adenoma. Hypothyroidism — dogs , bilateral symmetric alopecia, "tragic face", obesity, atherosclerosis
Parathyroid	Secondary hyperparathyroidism: nutritional (low Ca : high P, bran) → big head , and renal (chronic kidney disease) → rubber jaw . Both give fibrous osteodystrophy
Adrenal	Hyperadrenocorticism (Cushing) — dog, usually pituitary-dependent ; pot belly, symmetric alopecia, thin skin, calcinosis cutis , hepatomegaly with steroid hepatopathy, stress leucogram . Hypoadrenocorticism (Addison) — collapse, hyperkalaemia + hyponatraemia , Na:K < 27
Pancreatic islets	Diabetes mellitus — hyperglycaemia + glucosuria, ketonuria, hepatic lipidosis, cataract in dogs; islet amyloidosis in cats. Insulinoma — hypoglycaemia, seizures
Pituitary	Adenoma → Cushing; diabetes insipidus (ADH) — huge volumes of dilute urine, SG ~1.005

STRESS LEUCOGRAM ★ **Neutrophilia** · **lymphopenia** · **eosinopenia** · **monocytosis**. Remember it as the **steroid** pattern — the opposite of a parasitic pattern (eosinophilia).

21 • SKIN — THE VOCABULARY IS THE EXAM

Gross term	Meaning	Micro term	Meaning
Macule	Flat colour change	Hyperkeratosis	Thick keratin layer
Papule	Small solid raised	Parakeratosis	Nuclei retained in keratin — ZINC deficiency in pigs
Vesicle / bulla	Fluid-filled	Acanthosis	Thick stratum spinosum
Pustule	Pus-filled	Acantholysis	Cells fall apart — PEMPHIGUS
Wheal	Oedematous, transient	Spongiosis	Intercellular oedema
Scale / crust	Keratin / dried exudate	Ballooning degeneration	Swollen keratinocytes — POX viruses

Named skin lesions: **ringworm** (circular scaly alopecia, KOH mount + Wood's lamp) · **mange** (*Sarcoptes* intense pruritus, *Demodex* non-pruritic, skin scraping) · **zinc-responsive parakeratosis** of pigs · **hyperkeratosis** of chlorinated naphthalene poisoning and vitamin A deficiency · **photosensitization** on white skin · **papillomatosis** (warts, bovine papillomavirus) · **dermatophilosis** ("rain scald", "railroad-track" branching filaments) · **lumpy skin disease** nodules.

22 • EAR & EYE

- **Otitis externa** — *Otodectes* ear mite, *Malassezia*, bacteria; long pendulous ears predispose
- **Aural haematoma** — blood between cartilage and perichondrium, from head shaking
- **Otitis media/interna** — head tilt, extension of pharyngeal infection through the Eustachian tube
- **IBK "pinkeye" ★** — *Moraxella bovis*, fly-spread; keratoconjunctivitis, corneal ulcer → opacity, possible perforation
- **"BLUE EYE" ★** — **anterior uveitis with corneal oedema** after **infectious canine hepatitis** (CAV-1) or its vaccine; a **Type III** immune-complex reaction
- **Equine recurrent uveitis ("moon blindness")** — immune-mediated, associated with *Leptospira*; leads to cataract, synechia and phthisis bulbi
- **Squamous cell carcinoma of the eye** — **Hereford** and other unpigmented-eyelid cattle, sunlight; commonest bovine eye tumour
- **Cataract** — lens opacity (diabetes, age, trauma) · **Glaucoma** — ↑ intraocular pressure, buphthalmos · **Exophthalmos** vs **enophthalmos**

23 • BLANK-FILL DRILL — UNIT 2

- Wooden tongue = ***Actinobacillus lignieresii***; lumpy jaw = ***Actinomyces bovis***
- Antemortem bloat is proved by the **bloat line** in the oesophagus
- Rumenitis–liver abscess complex organism = ***Fusobacterium necrophorum***
- Johne's disease gives a **corrugated/rugose** ileum with **no ulceration**
- Button ulcers of the caecum occur in **classical swine fever**
- Bronchopneumonia is **cranioventral**; interstitial pneumonia is **diffuse**
- Marbling of the lung with sequestra = **CBPP**
- Lung that has breathed **floats** in water — the **docimasia** test
- Atrophic rhinitis of pigs is caused by ***Bordetella bronchiseptica*** + toxigenic ***Pasteurella multocida*** type D
- White muscle disease is deficiency of **vitamin E and selenium**
- In myoglobinuria the plasma stays **clear**; in haemoglobinuria it is **red**
- Rickets affects the **young**; osteomalacia the **adult**
- Big head of horses = **nutritional secondary hyperparathyroidism**; rubber jaw = **renal** secondary hyperparathyroidism
- Tiger heart is classic of **FMD** in calves
- Vegetative endocarditis affects the **right AV** valve in cattle and the **left AV (mitral)** valve in pigs
- Verminous aneurysm of the horse is caused by ***Strongylus vulgaris*** in the **cranial mesenteric** artery
- Microcytic hypochromic anaemia means **iron deficiency**
- The **horse** does not release reticulocytes, so its anaemia cannot be graded on a smear
- Onion-ring pus in a sheep lymph node = ***Corynebacterium pseudotuberculosis***
- Blackberry-jam spleen with unclotted tarry blood = **anthrax**; confirm with **McFadyean's** reaction
- Marginal wedge-shaped splenic infarcts = **classical swine fever**
- White-spotted kidney of calves = **interstitial nephritis**, often **leptospirosis**
- Pyelonephritis in cattle is caused by ***Corynebacterium renale***
- Ethylene glycol poisoning deposits **calcium oxalate** crystals in tubules
- Enzootic haematuria follows chronic feeding of **bracken fern**
- Necrotic cotyledons with thick leathery intercotyledonary membrane = **brucellosis**
- Pyometra of the bitch is driven by **progesterone** with ***E. coli***
- Gangrenous mastitis is usually caused by ***Staphylococcus aureus***
- Perivascular cuffing indicates a **viral** encephalitis
- Negri bodies are **intracytoplasmic** and lie in **hippocampal** neurons and **Purkinje** cells
- Polioencephalomalacia is due to **thiamine** deficiency and **autofluoresces** under UV light
- Leukoencephalomalacia of horses follows **mouldy maize (fumonisin)**
- Circling disease with brain-stem microabscesses = **listeriosis**
- Sway back of lambs is **copper** deficiency
- Cerebellar hypoplasia in calves follows in-utero **BVD** infection
- The stress leucogram is **neutrophilia, lymphopenia, eosinopenia, monocytosis**
- Parakeratosis in pigs indicates **zinc** deficiency
- Acantholysis is the hallmark of **pemphigus**; ballooning degeneration of **pox**
- Pinkeye of cattle is caused by ***Moraxella bovis***
- Blue eye after ICH vaccination is a type **III** hypersensitivity